

Worksheet 3

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Branch: MCA(General)

Semester: III

Subject Name: Internet Of Things

UID:25MCA20259

Section/Group:25MCA-1-A

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1. Aim/Overview of the practical:

Interface an Arduino Uno with Ultrasonic Sensor to calculate the distance and print on a Serial monitor and on LCD. Follow the mentioned cases after the calculation of distance.

- If the distance is > 250 cm green light is in ON condition.
- If the distance is < 250 cm and the distance > 90 yellow light is in ON condition and the green light remains Off
- If the distance is < 90 cm turn on the buzzer and lights will remain OFF.

2. Apparatus:

Hardware Requirements: Arduino Uno Board

Ultrasonic Sensor

LCD (16*2)

Software requirements: TinkerCAD

MS Windows 10

3. Circuit Diagram(TinkerCad): (Case 1)

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
// Ultrasonic sensor
const int trigPin = 9;
const int echoPin = 10;
```

```
// Green LED
const int greenLED = 6;
void setup(){
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(greenLED, OUTPUT);
```

```
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.print("Distance:");
}
```

void loop()

```
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
```

```
long duration = pulseIn(echoPin, HIGH);
float distance = duration * 0.0343 / 2;
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");
```

```
lcd.setCursor(0, 0);
lcd.print("Distance:   ");
lcd.setCursor(0, 1);
lcd.print(distance);
lcd.print(" cm   ");
```

```
if (distance > 250){
  digitalWrite(greenLED, HIGH);
  Serial.println("GREEN LED ON");
}
else{
  digitalWrite(greenLED, LOW);
}
delay(500);
```

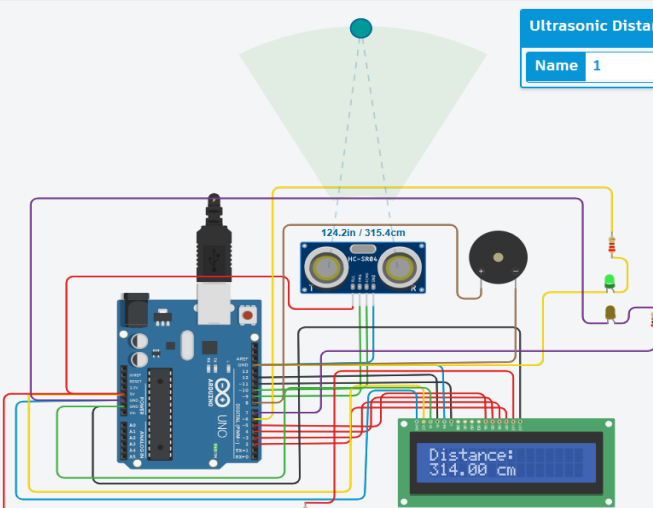
}

OUTPUT:

Anindita_Dhar_25MCA20259

Simulator time: 00:00:38

Ultrasonic Distance Sensor...
Name 1



How the debugger works
Add breakpoints by clicking on numbers. Hover over the variable to see their value. Use the buttons above to resume simulation step one line at a time.

```
20 pinMode(greenLED, OUTPUT);
21 pinMode(yellowLED, OUTPUT);
22 pinMode(buzzer, OUTPUT);
23
24 Serial.begin(9600);
25
26 delay(100);
27 lcd.begin(16, 2);
28 lcd.clear();
29 lcd.print("Distance:");
30 }
31
32 void loop()
33 {
34   // Send ultrasonic pulse
35   digitalWrite(trigPin, LOW);
36   delayMicroseconds(5);
37
38   digitalWrite(trigPin, HIGH);
39   delayMicroseconds(10);
40
41   // Receive echo pulse
42   long duration = pulseIn(echoPin, HIGH);
43   float distance = duration * 0.0343 / 2;
44   Serial.print("Distance: ");
45   Serial.print(distance);
46   Serial.println(" cm");
47
48   lcd.setCursor(0, 0);
49   lcd.print("Distance:   ");
50   lcd.setCursor(0, 1);
51   lcd.print(distance);
52   lcd.print(" cm   ");
53
54   if (distance > 250){
55     digitalWrite(greenLED, HIGH);
56     Serial.println("GREEN LED ON");
57   }
58   else{
59     digitalWrite(greenLED, LOW);
60   }
61   delay(500);
62 }
```

Serial Monitor

```
Distance: 314.10 cm
GREEN LED ON
Distance: 314.26 cm
GREEN LED ON
Distance: 314.10 cm
GREEN LED ON
Distance: 314
```

(Case 2)**Coding:**

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

// Ultrasonic sensor
const int trigPin = 9;
const int echoPin = 10;
// Yellow LED
const int yellowLED = 7;
void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(yellowLED, OUTPUT);
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.print("Distance:");
}

void loop()
{
  // Send ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Measure echo time
  long duration = pulseIn(echoPin, HIGH);

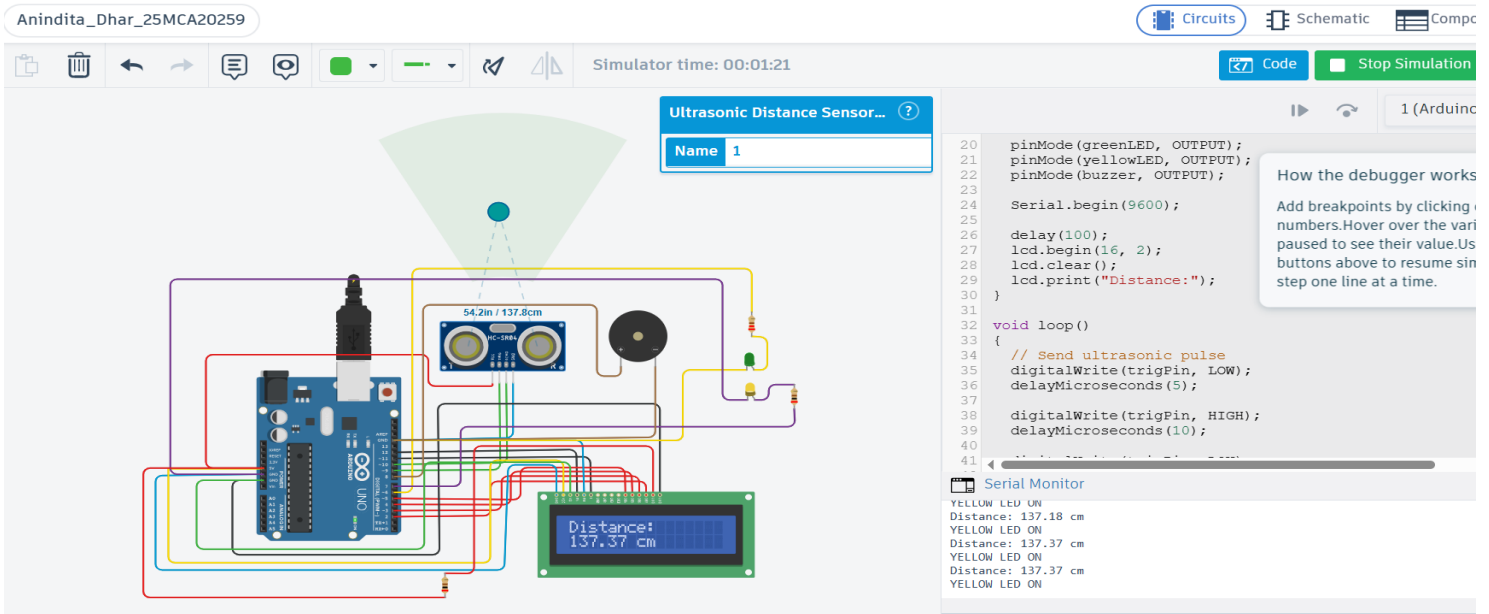
  // Calculate distance in cm
  float distance = duration * 0.0343 / 2;

  // Print on Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  // Print on LCD
  lcd.setCursor(0, 0);
  lcd.print("Distance:  ");
  lcd.setCursor(0, 1);
  lcd.print(distance);
  lcd.print(" cm  ");
```

```
if (distance < 250 && distance > 90)
{
    digitalWrite(yellowLED, HIGH);
    Serial.println("YELLOW LED ON");
}
else
{
    digitalWrite(yellowLED, LOW);
}
delay(500);
}
```

OUTPUT:



The screenshot shows an Arduino IDE simulation. The circuit includes an Arduino Uno, an ultrasonic sensor (HC-SR04), a yellow LED, a buzzer, and an LCD display. The code is as follows:

```
20 pinMode(greenLED, OUTPUT);
21 pinMode(yellowLED, OUTPUT);
22 pinMode(buzzer, OUTPUT);
23
24 Serial.begin(9600);
25
26 delay(100);
27 lcd.begin(16, 2);
28 lcd.clear();
29 lcd.print("Distance:");
30
31
32 void loop()
33 {
34     // Send ultrasonic pulse
35     digitalWrite(trigPin, LOW);
36     delayMicroseconds(5);
37     digitalWrite(trigPin, HIGH);
38     delayMicroseconds(10);
39
40
41 }
```

The serial monitor shows the following output:

```
YELLOW LED ON
Distance: 137.18 cm
YELLOW LED ON
Distance: 137.37 cm
YELLOW LED ON
Distance: 137.37 cm
YELLOW LED ON
```

(Case 3)

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
// Ultrasonic sensor
const int trigPin = 9;
const int echoPin = 10;
// Buzzer
const int buzzer = 8;

void setup()
{
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(buzzer, OUTPUT);
    Serial.begin(9600);
    lcd.begin(16, 2);
}
```

```

lcd.print("Distance:");
}

void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  float distance = duration * 0.0343 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  lcd.setCursor(0, 0);
  lcd.print("Distance:   ");
  lcd.setCursor(0, 1);
  lcd.print(distance);
  lcd.print(" cm   ");
  if (distance < 90)
  {
    digitalWrite(buzzer, HIGH);
    Serial.println("BUZZER ON");
  }
  else
  {
    digitalWrite(buzzer, LOW);
  }
  delay(500);
}

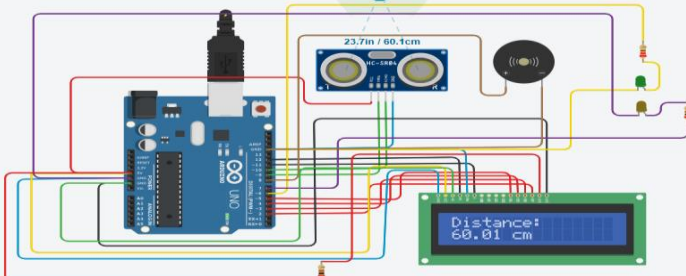
```

OUTPUT:

Anindita_Dhar_25MCA20259

Simulator time: 00:01:40

Ultrasonic Distance Sensor... ?
Name 1



20 pinMode(greenLED, OUTPUT);
21 pinMode(yellowLED, OUTPUT);
22 pinMode(buzzer, OUTPUT);
23
24 Serial.begin(9600);
25
26 delay(100);
27 lcd.begin(16, 2);
28 lcd.clear();
29 lcd.print("Distance:");
30
31
32 void loop()
33 {
34 // Send ultrasonic pulse
35 digitalWrite(trigPin, LOW);
36 delayMicroseconds(5);
37 digitalWrite(trigPin, HIGH);
38 delayMicroseconds(10);
39
40
41

How the debugg
Add breakpoints b
numbers. Hover ov
paused to see the
buttons above to r
step one line at a

Serial Monitor

```

BUZZER LW
Distance: 59.77 cm
BUZZER ON
Distance: 60.01 cm
BUZZER ON
Distance: 60.01 cm
BUZZER ON

```

4. Learning outcomes (What I have learnt):

1. Ultrasonic sensors and its properties.
2. Interfacing of ultrasonic sensors with Arduino Uno and LCD.
3. Connecting ultrasonic sensor ,Arduino Uno board and LCD in a circuit.